

Reason about equivalent expressions

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent? Copy or complete the first step.

2. Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?

3. Insert operations between 6, 5 and 4 to make 26.

4. Miro says: Miro thinks changing brackets never changes an answer. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent? Copy or complete the first step.

Answer Yes, both equal 161.

2. Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?

Answer $(100 - 8) \times 9$ is greater: 828 compared with 28.

3. Insert operations between 6, 5 and 4 to make 26.

Answer $6 \times 5 - 4 = 26$.

4. Miro says: Miro thinks changing brackets never changes an answer. Is this correct? Explain.

Answer Brackets can change which values combine first, producing a different result.

Reason about equivalent expressions

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent?

6. Check this result using an inverse, estimate or known fact: Insert operations between 6, 5 and 4 to make 26.

2. Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?

7. Find three expressions with answer 50 using at least two different operations.

3. Insert operations between 6, 5 and 4 to make 26.

8. Identify and correct this misconception: Miro thinks changing brackets never changes an answer.

4. Solve this independently and show every stage: Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?

10. Write and solve a similar question to: Insert operations between 6, 5 and 4 to make 26.

Teacher answers and checking prompts

1. Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent?
Answer Yes, both equal 161.
2. Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?
Answer $(100 - 8) \times 9$ is greater: 828 compared with 28.
3. Insert operations between 6, 5 and 4 to make 26.
Answer $6 \times 5 - 4 = 26$.
4. Solve this independently and show every stage: Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent?
Answer Yes, both equal 161.
5. Use a second method or representation for: Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?
Answer Expected result: $(100 - 8) \times 9$ is greater: 828 compared with 28. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Insert operations between 6, 5 and 4 to make 26.
Answer Expected result: $6 \times 5 - 4 = 26$. A valid check must be shown.
7. Find three expressions with answer 50 using at least two different operations.
Answer Answers vary, for example $8 \times 6 + 2$; $100 \div 2$; $(15 - 5) \times 5$.
8. Identify and correct this misconception: Miro thinks changing brackets never changes an answer.
Answer Brackets can change which values combine first, producing a different result.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Insert operations between 6, 5 and 4 to make 26.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reason about equivalent expressions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find three expressions with answer 50 using at least two different operations.

6. Create a new example that tests the same idea as: Insert operations between 6, 5 and 4 to make 26.

2. Prove the answer to this question, rather than only calculating it: Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $6 \times 5 - 4 = 26$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro thinks changing brackets never changes an answer.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find three expressions with answer 50 using at least two different operations.
Answer Answers vary, for example $8 \times 6 + 2$; $100 \div 2$; $(15 - 5) \times 5$.
2. Prove the answer to this question, rather than only calculating it: Are $7 \times (20 + 3)$ and $7 \times 20 + 7 \times 3$ equivalent?
Answer Expected result: Yes, both equal 161. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Which is greater: $100 - 8 \times 9$ or $(100 - 8) \times 9$?
Answer Expected result: $(100 - 8) \times 9$ is greater: 828 compared with 28. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $6 \times 5 - 4 = 26$.
Answer One valid reconstruction is: Insert operations between 6, 5 and 4 to make 26. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro thinks changing brackets never changes an answer.
Answer Brackets can change which values combine first, producing a different result.
6. Create a new example that tests the same idea as: Insert operations between 6, 5 and 4 to make 26.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.